

Student Guide — middle / module-14-magic-squares

IKS Curriculum

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Student Workbook

Student Workbook — Module 14 Magic Squares (Middle)

Name: _____ Class: _____ Section: _____

This is your workbook for the next two weeks. Bring it to every class.

Day 1 — What is a magic square?

Definition: A magic square is an $n \times n$ grid of distinct numbers where every row, every column, and both main diagonals add up to the same total — the **magic constant**.

My first 3×3 attempt:

—		—		—
—		—		—
—		—		—

Class 3×3 (verified):

2		7		6

9		5		1

4		3		8

Row sums: __ , __ , __ · Column sums: __ , __ , __ · Diagonal sums: __ , __

Day 2 — The magic constant

Formula: $S = n(n^2 + 1) / 2$

n	S
3	_____
4	_____
5	_____
6	_____
7	_____
8	_____

Day 3 — Counting and symmetry

There is only **ONE** 3×3 magic square with 1–9, **up to symmetry**.

Number of rotations + reflections = _____

Why must 5 be at the centre? (one sentence)

Day 4 — The stair-step (*turagagati*) method

Five steps: 1. Place 1 in _____. 2. From cell with k , move _____ for $k + 1$. 3. Off-top $\rightarrow$ wrap to _____. 4. Off-right $\rightarrow$ wrap to _____. 5. Blocked $\rightarrow$ place $k + 1$ _____.

My 3×3 by *turagagati*:

—		—		—
—		—		—
—		—		—

Day 5 — My 5×5 magic square

—		—		—		—		—
—		—		—		—		—
—		—		—		—		—
—		—		—		—		—
—		—		—		—		—

Magic constant should be: _____ · First-row sum: _____ · Main-diagonal sum: _____

Day 6 — Khajuraho

7		12		1		14
2		13		8		11
16		3		10		5
9		6		15		4

Magic constant: ____ · Top-left 2×2 sum: ____ · Centre 2×2 sum: ____

Notable: Khajuraho's square is *panel-magic* — every 2×2 panel also sums to 34.

Day 7 — Nārāyaṇa Paṇḍita

- Wrote *Gaṇita-kaumudī* in _____ CE
 - Book 14 is called _____ (“mathematics of auspicious squares”)
 - Why does having a *rule* matter more than having one *example*? (your paragraph)
-
-
-
-
-
-

Day 8 — Cross-cultural

	Lo Shu	Khajuraho	Dürer
Date	~5th c. BCE	~10th–11th c. CE	1514 CE
Order	3	4	4
Constant	_____	_____	_____
Panel-magic?	(n/a)	_____	_____

My project choice: _____ (poster / video / written exposition) **Partner:** _____

Day 9 — Sudoku vs magic square

Sudoku constrains _____; magic squares constrain _____.

Project progress today: _____

Day 10 — Reflection

1. What did you set out to make, and what did you end up with?

2. What was the most surprising thing you learned in this module?

3. If you had 3 more days, what would you do differently?

Glossary (your own)

Sanskrit	Meaning	First met
<i>Bhadragaṇita</i>	mathematics of auspicious squares	Day 1
<i>Gaṇita-kaumudī</i>	moonlight of mathematics	Day 1
<i>samānaṅka</i>	equal-sum (the magic constant)	Day 2
<i>turagagati</i>	horse-step / stair-step method	Day 4
<i>yantra</i>	diagram	Day 6

Day 1 — What is a Magic Square?

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) state the definition of a magic square, and (b) construct a 3×3 magic square with the numbers 1 to 9 by their own trial.

Materials

- Printed blank 3×3 grid (3 per student, with the digits 1–9 listed beside the first grid)
- A loose set of paper tiles or small cards numbered 1–9 (one set per pair)
- Whiteboard

Vocabulary introduced today

- *magic square* — a square grid in which every row, every column, and both main diagonals add up to the same total.
- *magic constant* — that common total (sometimes also called the *magic sum*).

Lesson flow

Warm-up (5 min)

- Teacher draws an empty 3×3 grid on the board. “*I want to place the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 in this grid — each number exactly once — so that **every row, every column, and both diagonals add up to the same number**. Can it even be done?*”
- Take 2–3 quick guesses. Don’t confirm. Write on the side of the board: “Possible? If yes — what’s the magic sum?”

Core (20 min)

1. **Hand out tiles 1–9 and the blank grids.** Pairs work for 8 minutes to try to build one.
2. **Mid-pair check-in (after 4 min):** “*What number is in the centre of your best attempt so far?*” Most pairs will have 5 in the centre. Ask why.
3. **Give a hint, not the answer.** “*The middle number is 5. Each row sums to 15. Now try again.*”
4. **Reveal one valid square** on the board (after most pairs have struggled — 7–8 min in):

2		7		6

9		5		1

4		3		8

Verify together: each row $\rightarrow$ 15. Each column $\rightarrow$ 15. Both diagonals $\rightarrow$ 15. “*That sum 15 is called the magic constant of this square.*”

5. **Definition on the board.** Students copy this into their notebooks:

A **magic square** is an $n \times n$ grid filled with distinct numbers such that every row, every column, and both main diagonals add up to the same total — the **magic constant**.

Activity (15 min) — Verify and vary

- **Part A (5 min):** Each student picks any 3 lines (rows, columns, or diagonals) of the revealed square and verifies the sum on their notebook.
- **Part B (10 min):** Pairs try to make a *different* 3×3 magic square using 1–9. (Without saying so, you are previewing Day 3 — they will discover symmetries.) Most will produce rotations or reflections of the same square. That is the discovery, not the failure.

Wrap-up (5 min)

- Quick show of hands: “Did anyone find a square that is genuinely different from the one on the board?” (Most will say yes; some of those “different” squares will be rotations.)
- Preview Day 2: “*Tomorrow we’ll prove why the magic constant is exactly 15 for the 3×3 with 1–9. We won’t guess — we’ll derive it.*”

Student-facing content

A **magic square** is an $n \times n$ grid of distinct numbers where every row, every column, and both main diagonals add up to the same total — the **magic constant**.

The simplest non-trivial magic square uses the numbers 1 to 9 in a 3×3 grid. Its magic constant is **15**.

2		7		6

9		5		1

4		3		8

Verify: rows $\rightarrow 2+7+6=15, 9+5+1=15, 4+3+8=15$. Columns $\rightarrow 2+9+4=15, 7+5+3=15, 6+1+8=15$. Diagonals $\rightarrow 2+5+8=15, 6+5+4=15$.

Homework

- On graph paper, copy the magic square shown in class. Then, beside it, draw three different “rotations” of it (turn the page 90° , 180° , 270°). Are they still magic squares? Verify one row of each.

Differentiation

- **One tier up:** Try to make a 4×4 magic square using the numbers 1 to 16. (Hard! The magic constant is 34. You will likely fail without a method. That’s fine — we’ll learn the method later.)
- **One tier down:** Practise just verifying the revealed 3×3 square. Sum every row, column, and diagonal slowly and carefully.

Teacher notes

- Resist the urge to give the answer in the first 5 minutes. The discovery is the lesson.
- Some students will ask “is it always 15?” Answer: “for THIS square, with THESE numbers (1–9). Tomorrow we’ll see why.”
- A few students will try to use the same number twice. Reinforce that all 9 numbers must be distinct — that’s part of the definition.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — the historical source for the systematic treatment of magic squares we will follow this module. Today we set the stage; Day 4 begins his explicit method.

Day 2 — The Magic Constant

Module: Magic Squares (*Bhadrakarita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) derive the magic constant $S = n(n^2 + 1)/2$ for the 3×3 case by direct argument, and (b) apply the formula to predict the constant for $n = 4, 5, 6, 7$.

Materials

- Whiteboard
- Notebook + pen
- The 3×3 magic square from Day 1, visible on the board

Vocabulary review

- *magic square, magic constant* — defined Day 1.

Lesson flow

Warm-up (5 min)

- Recall chant: “ 3×3 square with 1 to 9 — magic constant is ____?” Class: “15.”
- “Why exactly 15, though? Why not 14 or 17?” Take 2 student responses.

Core (20 min)

1. **The sum argument.** Write on the board:

Add up all the numbers in the square: $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45$.

“This total goes into the square once. Now — the square has 3 rows. Each row has the same sum S . So $3 \times S = 45 \rightarrow S = 15$.”

Repeat the logic with student input. They should arrive at the formula step-by-step.

2. **Generalise.** For an $n \times n$ square containing the numbers 1, 2, ..., n^2 :

Sum of all numbers = $1 + 2 + \dots + n^2 = n^2(n^2 + 1) / 2$

The square has n rows, each summing to S . So:

$$n \times S = n^2(n^2 + 1) / 2 \rightarrow \mathbf{S = n(n^2 + 1) / 2}$$

3. **Apply.** Compute together for several n :

n	n^2	n^2+1	$n(n^2+1)/2$
3	9	10	$3 \times 10 / 2 = \mathbf{15}$
4	16	17	$4 \times 17 / 2 = \mathbf{34}$
5	25	26	$5 \times 26 / 2 = \mathbf{65}$
6	36	37	$6 \times 37 / 2 = \mathbf{111}$

n	n^2	n^2+1	$n(n^2+1)/2$
7	49	50	$7 \times 50 / 2 = 175$

4. **A paraphrase from the source.** Nārāyaṇa Paṇḍita, in *Bhadragaṇita* (Book 14 of *Gaṇita-kaumudī*, 1356 CE), gives the magic-constant rule explicitly: the common sum is one n -th of the sum of the numbers in the square. We’ve just re-derived his rule. (See *sources.md*.)

Activity (15 min) — Formative quiz Part A

- Hand out quizzes/formative.md Part A.
- Students complete individually. 10 minutes; 3 minutes for self-check against the board answer key.
- Part A is 6 short questions: 3 on the formula, 3 on verifying given squares.

Wrap-up (5 min)

- “Quick check: what would the magic constant be for a 10×10 magic square with the numbers 1 to 100?” Class works it out: $S = 10 \times 101 / 2 = 505$.
- Preview Day 3: “Tomorrow we ask: how many different 3×3 magic squares are there with 1 to 9? Take guesses tonight.”

Student-facing content

The magic constant formula.

For an $n \times n$ magic square containing the numbers 1, 2, ..., n^2 :

$$S = n(n^2 + 1) / 2$$

Why? The sum of all n^2 numbers is $n^2(n^2 + 1) / 2$ (the standard sum-to- n formula). This total is split equally among the n rows, each of which sums to S . So $n \times S = n^2(n^2 + 1) / 2 \rightarrow$ divide both sides by n .

n	Constant
3	15
4	34
5	65
6	111
7	175

Nārāyaṇa Paṇḍita stated this rule in *Bhadragaṇita* (1356 CE).

Homework

- Verify the formula for $n = 8$ and $n = 9$. (Answers: 260 and 369.)
- For each, also compute the total sum $n^2(n^2 + 1)/2$ directly and confirm it equals $n \times S$.

Differentiation

- **One tier up:** What if the magic square contains *consecutive* numbers but starting at k , not at 1? (E.g. a 3×3 with 5, 6, 7, ..., 13.) Derive a formula for the new magic constant. (Answer: $S = n \times k + n(n^2-1)/2$.)
- **One tier down:** Focus on the 3×3 case alone. Verify the constant 15 by summing every row, column, and diagonal of the Day-1 square. Do it slowly.

Teacher notes

- A common stumble: students compute $1 + 2 + \dots + 9$ incorrectly. Remind them that $1+9$, $2+8$, $3+7$, $4+6$ each = 10, plus 5 = 45. This itself is a pretty argument worth pausing on.
- For the generalisation, some students will struggle with the n^2 notation. Walk them through the $n = 5$ case explicitly: $1 + 2 + \dots + 25 = 25 \times 26 / 2 = 325$; split among 5 rows $\rightarrow 65$.
- The formula $S = n(n^2 + 1)/2$ is **the** rule of this module. Drill it.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — paraphrased here: the magic constant equals one n -th of the total sum of the n^2 entries.

Day 3 — How Many 3×3 Magic Squares Are There?

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) discover that the 3×3 magic square with 1–9 is **unique up to symmetry**, (b) list the 8 symmetries (4 rotations + 4 reflections), and (c) explain why 5 must occupy the centre.

Materials

- Printed 3×3 grids (5 per pair) with the digits 1–9 listed
- Coloured pencils (3 colours per student)
- The Day-1 square visible on the board

Vocabulary introduced today

- *symmetry* — a way of moving or flipping a figure so that it looks the same in a different orientation.
- *rotation* — turning around a centre by 90° , 180° , or 270° .
- *reflection* — flipping across a line (horizontal, vertical, or one of the diagonals).

Lesson flow

Warm-up (5 min)

- Daily fluency: “row sums to 15 — give me the missing number.” Quick 5 rounds.
- Recall: “Yesterday we proved the magic constant for a 3×3 is 15. Today’s question: how many different 3×3 magic squares with 1–9 are there?”
- Take 3 student guesses. Write the guesses on the board.

Core (20 min)

1. **Why 5 must be in the centre.** Argue together:

- The centre cell sits on 4 lines: the middle row, the middle column, and both diagonals.
- The four corner cells each sit on 3 lines (a row, a column, and one diagonal).
- The four edge-middle cells each sit on 2 lines (a row and a column).
- Sum of all lines = $4 \times 15 = 60$ (4 lines through the centre? — actually 8 lines total in a 3×3 : 3 rows + 3 columns + 2 diagonals. Their sum is $8 \times 15 = 120$.)
- In that sum of 120, each cell is counted as many times as the number of lines it sits on: centre counted 4 times, corners 3 times each, edges 2 times each.
- Total: $4c + 3(\text{corner sum}) + 2(\text{edge sum}) = 120$, where c is the centre value.
- Corner sum + edge sum + $c = 45$ (the total of 1..9). So $3(\text{corner+edge}) + 4c + \dots$ — work it out together. The cleanest answer: $c = 5$.

(Don’t make them complete the full algebra — show the structure, conclude $5 = \text{centre}$.)

2. **Why 2, 4, 6, 8 must be at corners.** The four corner cells must contain the four *even* numbers from 1–9 (which are 2, 4, 6, 8). Reason: corners sit on diagonals — and the diagonals also sum to 15. With 5 in the centre, each diagonal pair must sum to 10. Pairs from 1–9 (excluding 5) summing to 10 are (1,9), (2,8), (3,7), (4,6). The two diagonals use four of these — pick any two pairs. The remaining four numbers (the odd or even ones, depending on the pair choice) end up at the edges. Most cleanly: even numbers at corners.

3. **The 8 symmetries.** Demonstrate on the board. Take the Day-1 square and:

- Rotate 90° , 180° , 270° — three new arrangements.
- Reflect horizontally, vertically, and across each diagonal — four more.
- Total: 8 different-looking arrangements, all the *same* magic square up to symmetry.

Theorem (state, don’t prove): There is exactly **ONE** 3×3 magic square with 1–9, *up to symmetry*. Counting all 8 rotated/reflected versions, you get 8 grids that all share the same essential structure.

Activity (15 min) — Find all 8

- Each pair starts with the Day-1 square. They must produce all 8 rotated/reflected versions on their own paper.

- Verify each by checking one row sums to 15.
- Pair share: any two pairs swap, and compare — did they get the same 8?

Wrap-up (5 min)

- “There is essentially *ONE* 3×3 magic square with 1–9. Eight different orientations of the same square. Tomorrow we ask: is there a method to construct magic squares of any size — not just 3×3 ?”
- Preview Day 4: Nārāyaṇa’s stair-step method.

Student-facing content

There is only ONE 3×3 magic square with 1–9 — up to symmetry.

The 4 rotations and 4 reflections together produce 8 grid arrangements. They all share the same structure: - 5 is in the centre. - The four even numbers (2, 4, 6, 8) are at the four corners. - The four odd numbers other than 5 (1, 3, 7, 9) are at the four edge-middles.

2 7 6	6 7 2	← reflection (horizontal flip)
-----	-----	
9 5 1	1 5 9	
-----	-----	
4 3 8	8 3 4	

Eight arrangements; one square.

Homework

- Take the 3×3 magic square and ask: if I change every entry by adding 10 (so the square becomes 12, 17, 16, 19, 15, 11, 14, 13, 18), is it still a magic square? What’s the new magic constant? Verify one row.

Differentiation

- **One tier up:** Try the same question for 4×4 magic squares — how many are there? Look it up (or estimate). The answer is 880 distinct 4×4 squares up to symmetry; that’s a surprise.
- **One tier down:** Just memorise: 5 in the centre, even at corners. Use coloured pencils to highlight the pattern in the Day-1 square.

Teacher notes

- The full proof that 5 must be central is a beautiful counting argument but takes 10 minutes if done thoroughly. Choose: either do it well, or do it briefly with hand-waves. Don’t do it half-fast — that confuses more than it clarifies.
- The “eight symmetries form a group” idea is the *dihedral group* D_4 . Don’t introduce that name at Middle band — note it for yourself; it returns at Senior band.
- A few students will produce a “9th” square that turns out to be one of the 8 in disguise. Walk them through which symmetry transforms theirs into the original.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — discusses equivalence classes of magic squares (paraphrased).

Day 4 — The Stair-Step Method (*turagagati*)

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will reproduce Nārāyaṇa Paṇḍita’s stair-step (*turagagati*) algorithm to construct a 3×3 magic square from scratch, and will be able to state all five steps of the rule.

Materials

- Whiteboard
- Printed blank 3×3 grids (3 per student)
- A printed copy of the five-step rule (handout) — see Student-facing content below

Vocabulary introduced today

- *turagagati* (IAST: turaga-gati; lit. “horse-step” or “knight-move”) — Nārāyaṇa Paṇḍita’s name for the diagonally-upward-right placement rule for constructing odd-order magic squares.

Lesson flow

Warm-up (5 min)

- Daily fluency: row-sum-to-15 quick fire. 5 rounds.
- “You’ve all built the 3×3 by guessing. Today we replace guessing with a method — Nārāyaṇa Paṇḍita’s *turagagati* rule from 1356 CE.”

Core (20 min)

1. **State the rule.** Five steps. Write them on the board. Have students copy.

The stair-step (*turagagati*) method — for any odd-order $n \times n$ square:

1. Place **1** in the **top-middle** cell.
2. From the cell containing k , move **one up and one to the right** (diagonally up-right) to place $k + 1$.
3. If you go off the **top edge**, wrap to the **bottom row** (same column you’d have landed in).
4. If you go off the **right edge**, wrap to the **left column** (same row).
5. If the destination cell is **already filled**, place $k + 1$ in the cell **directly below** the cell containing k instead.

Continue until all n^2 cells are filled.

2. **Demonstrate live on the 3×3.** Walk through cell-by-cell, narrating each step:

Step: 1	2	3	4	5
Cell: (1, c2)	(off-top,	(off-right,	filled –	(off-
top,	wrap to	wrap to	go directly	wrap
to	bottom)	left col)	below)	
bottom)				

The result:

. 1 .	. 1 .	. 1 .	. 1 .
2 1 .			
-----	-----	-----	
. . .	. . .	3 . .	3 . .
→ 3 . .			
-----	-----	-----	
. . .	. . 2	. . 2	4 . 2
4 . 2			

Continue: 6 (up-right from 5, wraps), 7 (up-right from 6, blocked → below), 8, 9. Final:

2 7 6

9 5 1

4 3 8

“Notice — this is exactly the square we discovered on Day 1. The method gave it to us directly, no guessing.”

3. **Why does this work?** Brief preview: the method spreads numbers evenly across rows, columns, and diagonals in a structured way. The full *proof* is Senior-band material. For now, the algorithm produces a magic square; verify by summing.

Activity (15 min) — Drill

- Use `activities/activity-01-stair-step-drill.md`.
- Each student constructs three 3×3 squares from scratch — first slowly, then faster.
- Pair check: did your neighbour get the same square? Verify magic constant = 15.

Wrap-up (5 min)

- Quick recap of the 5-step rule.
- Preview Day 5: *“Tomorrow we apply the same algorithm to a 5×5. The rule does not change.”*

Student-facing content

The stair-step (*turagagati*) method — for any odd-order $n \times n$ magic square:

1. Place **1** in the **top-middle** cell.
2. From the cell with k , move **one up and one to the right** to place $k + 1$.
3. If you go **off the top**, wrap to the **bottom row**.
4. If you go **off the right**, wrap to the **left column**.
5. If the destination is **already filled**, place $k + 1$ **directly below** the cell containing k instead.

2		7		6

9		5		1

4		3		8

Nārāyaṇa Paṇḍita described this method in *Bhadragaṇita* (1356 CE). Modern textbooks sometimes call it the *Siamese method* or *de la Loubère method* — named for a European traveller who reported it from Siam in 1687. Nārāyaṇa described it 300+ years earlier.

Homework

- Construct three 5×5 magic squares using the stair-step method. The magic constant should be 65 for all three. Yes — they will all be the same square. We'll discuss why tomorrow.

Differentiation

- **One tier up:** Try the method starting at the *bottom-middle* with the number 1 going *down-right* instead of up-right. Does it still produce a magic square? (Yes — a rotated version. Symmetry strikes again.)
- **One tier down:** Re-do the 3×3 construction with the rule sheet open in front of you. Speed comes later.

Teacher notes

- The most common student error is step 5 — when blocked, students go in random directions instead of “directly below.” Drill this. Repeat: “blocked → directly below.”
- A second common error: students try to apply this method to a 4×4 square. Remind them this is **odd-order only**. The 4×4 needs a different method (Day 6).
- When students wrap (step 3 or 4), they often “lose” themselves on the grid. A useful trick: imagine the grid tiled — the cell at (off-top, column c) is the same as (bottom row, column c).
- Walk around during the activity and correct misplacements early; mistakes compound.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — primary source for the *turagagati* method we are using.

- Plofker (2009), *Mathematics in India* — secondary scholarly context.

Day 5 — A 5×5 Magic Square

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will independently construct a 5×5 magic square using the *turagagati* method and verify its magic constant of 65.

Materials

- Printed blank 5×5 grids (3 per student)
- The 5-step rule handout from Day 4
- Stopwatch (for timed second-construction)

Vocabulary review

- *turagagati* — stair-step method (introduced Day 4).
- *magic constant* for 5×5 → **65** (from Day 2).

Lesson flow

Warm-up (5 min)

- Class recall: 5-step rule, recited together.
- Magic-constant check: “For a 5×5 with numbers 1–25, the magic constant is ____?” (65.)
- Pre-class question: did everyone produce the *same* 5×5 last night? Most will say yes.

Core (20 min)

1. **Build a 5×5 together on the board.** Narrate every step. Take 8–10 minutes — go cell-by-cell.

Start with 1 in row-1, column-3 (the top middle). Place 2 up-right (off the top → wraps to bottom row, column 4). Place 3 up-right from 2 (off the right → wraps to row 4, column 5)... continue.

Expected result (the classical Nārāyaṇa 5×5):

17		24		1		8		15

23		5		7		14		16

4		6		13		20		22

10		12		19		21		3

 11 | 18 | 25 | 2 | 9

2. **Verify together.** Pick:

- One row → $17 + 24 + 1 + 8 + 15 = 65$ ✓
- One column → $17 + 23 + 4 + 10 + 11 = 65$ ✓
- The main diagonal → $17 + 5 + 13 + 21 + 9 = 65$ ✓
- The anti-diagonal → $15 + 14 + 13 + 12 + 11 = 65$ ✓

3. **Notice the structure.** Take 3 minutes to observe:

- 13 (the middle number) is in the centre — consistent with the $n^2/2 + 1 = 13$ rule for odd-order.
- Numbers 1–9 occupy a “broken diagonal” pattern; 16–25 occupy the complementary pattern.
- The same algorithm that gave us the 3×3 gave us this. The rule does not change with size.

Activity (15 min) — Formative quiz Part B + speed construction

- **First 10 min:** Hand out quizzes/formative.md Part B. Students complete individually.
- **Last 5 min:** Each student starts a fresh 5×5 grid. On “Go,” they construct a magic square from scratch as fast as they can. Self-time. Verify one row before stopping the watch.

Wrap-up (5 min)

- Show of hands: who finished in under 4 minutes? Under 6? Under 8?
- “Yesterday’s home practice was the same 5×5. Tomorrow we look at a different beast — the 4×4.”
- Preview Day 6: Khajuraho and the 4×4.

Student-facing content

The 5×5 magic square — classical Nārāyaṇa form

Magic constant: 65 (from $S = n(n^2 + 1)/2$ with $n = 5$).

```

17 | 24 | 1 | 8 | 15
-----
23 | 5 | 7 | 14 | 16
-----
4 | 6 | 13 | 20 | 22
-----
10 | 12 | 19 | 21 | 3
-----
11 | 18 | 25 | 2 | 9

```

Built using the *turagagati* method, exactly as for the 3×3 — the rule does not change with size, as long as n is odd.

Homework

- Construct a 7×7 magic square using the stair-step method. The magic constant should be 175. Yes, it takes time — work patiently. Bring it tomorrow for spot-checks.

Differentiation

- **One tier up:** Construct a 9×9 magic square. Constant = 369. Time it.
- **One tier down:** Construct another 5×5 , slowly, with the rule sheet visible. Speed is not the goal yet.

Teacher notes

- For the live demonstration, use a slow pace and check student-following frequently — “where would 2 go?” “is that cell filled?”
- Students may resist because “I already built one at home.” Push them to *re-build* it on the board — narrating the rule reinforces it.
- A few students will produce mirror-image squares because they reversed the up-right direction. Show that mirror images are also valid magic squares (by symmetry from Day 3) — but consistency matters when you’re describing the algorithm.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadraṇita*), 1356 CE — the 5×5 example is one of the worked examples in the original text.

Day 6 — The 4×4 Square and Khajuraho

Module: Magic Squares (*Bhadraṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) recognise the Khajuraho 4×4 magic square, (b) verify its rows, columns, diagonals, and 2×2 panels all sum to 34, and (c) explain why this “panel-magic” property is *stronger* than the basic magic-square property.

Materials

- Printed image of the Khajuraho inscription panel (or a clean diagram of the square)
- Printed copy of the Khajuraho 4×4 grid (one per student) for sum-checking
- Coloured pencils

Vocabulary introduced today

- *Khajuraho* — site in Madhya Pradesh, India, with a temple complex dating to roughly the 10th–11th c. CE (Chandela dynasty). One temple wall carries a numbered 4×4 grid inscription.

- *panel-magic square* (also called *pandiagonal* or *most-perfect* in modern terminology) — a magic square where, in addition to rows, columns, and main diagonals, **every 2×2 panel** sums to the magic constant, and so do the **broken diagonals**.

Lesson flow

Warm-up (5 min)

- “Yesterday you built a 5×5 from scratch. The same algorithm doesn’t work for 4×4 — odd-order only. Today we look at one famous 4×4 — from a temple wall in central India, roughly 10th–11th c. CE.”
- Show the Khajuraho image. Don’t analyse yet. Just let students look.

Core (20 min)

1. **The Khajuraho 4×4.** Project or hand out:

7		12		1		14

2		13		8		11

16		3		10		5

9		6		15		4

“Magic constant should be 34. Let’s verify.” Walk through together:

- Row 1: $7 + 12 + 1 + 14 = 34$ ✓
 - Row 2: $2 + 13 + 8 + 11 = 34$ ✓
 - Rows 3, 4 — students verify independently.
 - All four columns — students verify in pairs.
 - Both main diagonals: $7+13+10+4 = 34$ ✓; $14+8+3+9 = 34$ ✓.
2. **The “extra magic” — panel sums.** “Try any 2×2 block. Does it also sum to 34?”

Pick a few:

- Top-left 2×2: $7 + 12 + 2 + 13 = 34$ ✓
- Top-right 2×2: $1 + 14 + 8 + 11 = 34$ ✓
- Centre 2×2: $13 + 8 + 3 + 10 = 34$ ✓
- Any 2×2, including those that “wrap” around the edge: also 34.

“This is a stronger property than just being a magic square. It is called **panel-magic** (or most-perfect or pandiagonal). Not every magic square has this.”

3. **A historical note.** The Khajuraho square is a real artefact — carved into a temple wall during the Chandela dynasty (~10th–11th c. CE). It is one of the earliest dated panel-magic 4×4 squares we have evidence of, anywhere. It predates Nārāyaṇa Paṇḍita (1356 CE) by several centuries; Nārāyaṇa later gave systematic methods for constructing such squares.

Honest framing: the exact dating of the Khajuraho inscription is debated among historians, but most place it in the 10th–11th century CE. The Chandela dynasty built the temple complex roughly between 950 and 1050 CE.

4. **The 4×4 construction (preview, not drilled).** The 4×4 cannot be built with the *turagagati* method. Nārāyaṇa’s *Bhadraṇita* gives a different recipe for orders divisible by 4 (called the *complement method*):

- Fill the 4×4 with 1–16 in natural order.
- Mark the cells on both main diagonals.
- For each *unmarked* cell, replace its value v with $17 - v$ (its complement to 17).

Demonstrate this on the board:

Natural-order fill:

1		2		3		4

5		6		7		8

9		10		11		12

13		14		15		16

After complementing unmarked cells:

1		15		14		4

12		6		7		9

8		10		11		5

13		3		2		16

Verify one row, one column, one diagonal — all 34. (*Note: this is a standard magic square, but not the same as the Khajuraho one, and not all such are panel-magic.*)

Activity (15 min) — Discussion + verification

- Pairs verify all eight 2×2 panels of the Khajuraho square — including the four “corner panels” formed by wrapping (top row joining bottom row, etc.).
- Class discussion: “*Why is the Khajuraho square called ‘most-perfect’?*” — student answers. Note their wording. The teacher’s frame: more constraints satisfied = more impressive.

Wrap-up (5 min)

- “Two takeaways. One: the 4×4 is built differently from the 5×5. Two: not all magic squares are equal — Khajuraho’s is unusually strong.”
- Preview Day 7: “Tomorrow we look at the mathematician who wrote down the methods: Nārāyaṇa Paṇḍita.”

Student-facing content

The Khajuraho 4×4 magic square — inscribed on a temple wall in Madhya Pradesh, ~10th–11th c. CE.

7		12		1		14

2		13		8		11

16		3		10		5

9		6		15		4

Magic constant: 34.

Extra property: every 2×2 panel also sums to 34. So do the “broken diagonals” (diagonals that wrap around the edges). This is called the **panel-magic** or **most-perfect** property — a much stronger constraint than the basic magic-square definition.

Nārāyaṇa Paṇḍita, in *Bhadragaṇita* (Book 14 of *Gaṇita-kaumudī*, 1356 CE), gave systematic methods for constructing such squares centuries later — though the Khajuraho artefact itself predates his treatise by 300+ years.

Homework

- Pick any 2×2 panel of the Khajuraho square. Verify the sum is 34. Now find a 2×2 panel that *wraps around* the edge (uses cells from both the top row and the bottom row). Verify that too. Write down both.

Differentiation

- **One tier up:** Find a 4×4 magic square that is *not* panel-magic — i.e. its rows/columns/diagonals work but some 2×2 panel does not equal 34. (Hint: the complement-method result given above.)
- **One tier down:** Just verify the basic property — rows, columns, diagonals. The panel property can come later.

Teacher notes

- Show the Khajuraho image if at all possible. Even a low-resolution photo. The artefact makes the lesson land.
- Don’t oversell the dating. “10th–11th century” is the conservative estimate; some sources claim earlier. Use the conservative range.
- The complement method shown is one of several 4×4 construction methods. Don’t drill it — Middle band only needs to know that *a* method exists.
- A few students will ask “is Sudoku a magic square?” Hold that question for Day 9. Short answer: no — Sudoku constrains rows/columns/boxes but not diagonals or sums.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — for the 4×4 construction methods.

- Material evidence: Khajuraho temple inscription (Madhya Pradesh, India), Chandela period (~10th–11th c. CE).

Day 7 — Nārāyaṇa Paṇḍita and *Gaṇita-kaumudī*

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) place Nārāyaṇa Paṇḍita (14th c. CE) in the chronology of Indian mathematics, (b) describe what the *Gaṇita-kaumudī* covers, and (c) write a short reflection on why systematic written method matters.

Materials

- Whiteboard
- Notebook
- A printed simple chronology chart (handout, see Student-facing content)

Vocabulary review

- *Gaṇita-kaumudī* (introduced in `_shared/glossary.md`) — “Moonlight of Mathematics,” Nārāyaṇa’s 1356 CE treatise.
- *Bhadragaṇita* — Book 14 of the *Gaṇita-kaumudī*, devoted to magic squares.

Lesson flow

Warm-up (5 min)

- “You’ve been using Nārāyaṇa Paṇḍita’s algorithm for three days. Today: who was he, what did he write, and why does it matter?”
- Quick free-write (60 seconds): “What do I think I know about Nārāyaṇa already?” — students jot. Not collected.

Core (20 min)

1. **Chronology.** Build a simple timeline on the board:

Period	Indian mathematics
~500 BCE	<i>Śulba Sūtras</i> — geometric ritual texts
~500 CE	Āryabhaṭa (b. 476) — <i>Āryabhaṭīya</i>
~600 CE	Brahmagupta — <i>Brāhmasphuṭasiddhānta</i>
~1100 CE	Bhāskara II — <i>Līlāvati</i> , <i>Bījagaṇita</i>
1356 CE	Nārāyaṇa Paṇḍita — <i>Gaṇita-kaumudī</i>
~1400 CE	Mādhava (Kerala) — early infinite

Period	Indian mathematics series
--------	---------------------------

“Nārāyaṇa wrote in the 14th century — relatively ‘late’ in the medieval Indian mathematical tradition, but he produced one of the most comprehensive single works.”

2. **What is in the *Gaṇita-kaumudī*?** Fourteen books (*vyavahāras*):

- Book 1–3: arithmetic (basic operations, fractions)
- Book 4–6: ratios, percentage problems, alligation
- Book 7–8: progressions and series
- Book 9: combinatorics (permutations, partitions — Nārāyaṇa was a pioneer here)
- Book 10–11: geometry of plane figures and solids
- Book 12: triangular and figurate numbers
- Book 13: practical problems (e.g. dilution of milk, mixture problems)
- **Book 14: *Bhadragaṇita* — magic squares**

“The treatise is comprehensive — it isn’t only about magic squares. But the magic-square book is the longest and most systematic in any pre-modern work we have.”

3. **Why systematic written method matters.** Discuss together:

- Before Nārāyaṇa, magic squares appeared in many Indian and non-Indian sources — but mostly as *examples*, not as recipes.
- Nārāyaṇa gives **methods**: rules that work for any input size. This is what makes it mathematics, not just a curiosity collection.
- This is the same distinction we draw today between “here’s a Sudoku puzzle” and “here’s an algorithm to generate Sudoku puzzles.”

4. **A paraphrase from the source.** From *Bhadragaṇita*, paraphrased:

“Magic squares of odd order are made by placing the first number in the top middle cell, and proceeding diagonally upward and rightward; if blocked, descend by one.”
(Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* Book 14, 1356 CE — paraphrased.)

This is the rule you have been using. The original is in Sanskrit; this is a faithful paraphrase, not a quoted verse.

Activity (15 min) — Reflection paragraph

- Each student writes 5–7 sentences in response to:
“Nārāyaṇa Paṇḍita gave us a rule, not a magic square. Why is having a rule more valuable than having just one example? Use specific evidence from this module.”
- Pairs swap and read each other’s paragraphs. One quick comment each: “the part that landed” and “what I’d want more of.”

Wrap-up (5 min)

- Two volunteer-read paragraphs (with permission).

- Preview Day 8: “*Tomorrow we go cross-cultural — China’s Lo Shu and Europe’s Dürer.*”

Student-facing content

Nārāyaṇa Paṇḍita (14th c. CE) was a mathematician of medieval India who wrote the *Gaṇita-kaumudī* (“Moonlight of Mathematics”) in **1356 CE**. The work has 14 books (*vyavahāras*) covering arithmetic, algebra, geometry, combinatorics, and recreational mathematics.

The 14th and final book is *Bhadragaṇita* (“Mathematics of Auspicious Squares”). It is the most comprehensive pre-modern treatment of magic squares in any tradition, giving:

- The magic-constant formula $S = n(n^2 + 1)/2$ (which you derived on Day 2).
- The stair-step (*turagagati*) method for any odd-order square (which you used on Days 4–5).
- Complement-based methods for even-order squares.
- Methods for *singly-even* orders (6, 10, 14, ...).
- Discussion of equivalence and symmetry classes.

Paraphrase from the source: “Magic squares of odd order are made by placing the first number in the top middle cell, and proceeding diagonally upward and rightward; if blocked, descend by one.” (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* Book 14, 1356 CE — paraphrased.)

Having a *rule* is more powerful than having an *example*. A rule generates infinitely many squares. An example gives you one.

Homework

- Polish your reflection paragraph from class. Add one more specific example (e.g. “I used the rule today to construct a 7×7 square in under 5 minutes; without the rule I would have been guessing all night.”).

Differentiation

- **One tier up:** Compare Nārāyaṇa to Bhāskara II (~1150 CE). What did Bhāskara cover that Nārāyaṇa didn’t? Vice versa? (Bhāskara: calculus precursors, planetary motion. Nārāyaṇa: combinatorics, magic squares.)
- **One tier down:** Write just 3–4 sentences. Focus on the “rule vs. example” distinction.

Teacher notes

- Avoid hero-worship framing. Nārāyaṇa was a brilliant systematiser; he was not the sole discoverer of magic squares.
- A few students will ask whether Nārāyaṇa’s methods were known outside India. Answer: yes, eventually — the same odd-order method shows up in 17th-c. European mathematics (via de la Loubère, 1687) probably with intermediate Islamic-world transmission. The independent re-discovery story is interesting but speculative.

- The reflection paragraph is a low-stakes formative — don't grade harshly. Read for understanding, not for polish.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — primary source paraphrased.
- Plofker (2009), *Mathematics in India*, ch. on 14th-c. mathematics — secondary scholarly context.

Day 8 — Cross-cultural: Lo Shu and Dürer

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) identify the Chinese *Lo Shu* and Albrecht Dürer's *Melencolia I* magic squares, (b) note structural similarities and differences across cultures, and (c) begin their project planning.

Materials

- Image of the Lo Shu square (turtle-shell pattern is helpful)
- Image of Dürer's *Melencolia I* (1514) with the 4×4 corner visible
- Project planning sheet (one per pair)

Vocabulary review

- *Lo Shu* — a Chinese 3×3 magic square attested at least from the *Da Dai Liji* (Han dynasty, ~5th–1st c. BCE) and probably earlier.
- *Melencolia I* — a 1514 engraving by the German artist Albrecht Dürer; its upper-right corner contains a famous 4×4 magic square.

Lesson flow

Warm-up (5 min)

- “*Magic squares are not a one-culture phenomenon. Today we look at two famous non-Indian examples and compare.*”
- Show both images side by side, briefly. Don't analyse yet.

Core (20 min)

1. The Lo Shu (China).

The 3×3 magic square in Chinese tradition is identical (up to symmetry) to the 3×3 we've been studying:

4 | 9 | 2

3		5		7

8		1		6

Magic constant: 15 (same as our Day-1 square).

The legend: Emperor Yu (legendary, ~2000 BCE attribution) saw the pattern on the shell of a turtle emerging from the Lo river. The earliest *documented* appearance in a Chinese text is the *Da Dai Liji* (Han dynasty, roughly 5th–1st c. BCE). Treat the legend as a story; the documented record is the historical evidence.

“This is the same square — up to symmetry — as the one we built on Day 1. Two cultures arrived at the same arrangement independently, separated by huge distances.”

2. Dürer’s Melencolia I (Germany, 1514 CE).

The 4×4 in Dürer’s engraving:

16		3		2		13

5		10		11		8

9		6		7		12

4		15		14		1

Magic constant: 34.

Notable features:

- The bottom-middle two cells contain 15 and 14 — encoding the date **1514**.
- The four corners (16, 13, 4, 1) sum to 34.
- Each 2×2 panel sums to 34 (panel-magic property — same as Khajuraho!).
- Dürer’s square is not the same as Khajuraho’s, but shares the panel-magic property.

3. Compare and contrast. Build a class table:

Square	Approx. date	Culture	Order	Magic constant	Special features
Lo Shu	~5th–1st c. BCE (documented)	Chinese	3×3	15	Identical to our 3×3
Khajuraho	~10th–11th c. CE	Indian	4×4	34	Panel-magic
Nārāyaṇa’s worked examples	1356 CE	Indian	up to large n	varies	Systematic methods

Square	Approx. date	Culture	Order	Magic constant	Special features
Dürer	1514 CE	German	4×4	34	Panel-magic; encodes year

4. **Honest framing.** Magic squares appeared independently (or via transmission) in:

- **China:** Lo Shu and later expansions.
- **India:** numerous artefacts and Nārāyaṇa’s systematic methods.
- **Islamic world:** treatises by al-Bīrūnī (~11th c. CE) and others.
- **Europe:** through Islamic and (likely) Indian transmission by 14th–15th c.

We do not have a clean story of “X invented it; Y borrowed it.” Frame the topic as a *cross-cultural mathematical fascination*.

Activity (15 min) — Project planning

- Use `activities/activity-02-cross-cultural-compare.md` (a side-by-side worksheet).
- Pairs commit to their project format (poster / video / written exposition) and one or two key ideas.
- Each pair writes a 3-sentence project plan, hands in for teacher review by end of class.

Wrap-up (5 min)

- Two-pair quick share: “Our project is going to be ____ because ____.”
- Preview Day 9: project work + a brief look at Sudoku.

Student-facing content

Magic squares across cultures.

Lo Shu (China, ~5th c. BCE)

```
4 | 9 | 2
-----
3 | 5 | 7
-----
8 | 1 | 6
```

Dürer (Germany, 1514 CE)

```
16 | 3 | 2 | 13
-----
5 | 10 | 11 | 8
-----
9 | 6 | 7 | 12
-----
4 | 15 | 14 | 1
```

The Lo Shu 3×3 is the same square as our Day-1 example (up to symmetry). The Dürer 4×4 shares the **panel-magic** property with the Khajuraho square (~10th–11th c. CE) — every 2×2 panel sums to 34.

Multiple cultures arrived at magic squares: China (~5th c. BCE earliest documented), India (Khajuraho ~10th c. CE, Nārāyaṇa’s *Bhadrakaṇṭa* 1356 CE), the Islamic world (~11th c. CE on), and Europe (~14th c. CE on). No clean single origin.

Homework

- Choose one of the three squares we've seen this fortnight (Day-1 / Khajuraho / Dürer). Verify its magic constant on a fresh piece of paper from scratch. Note one feature you find interesting.

Differentiation

- **One tier up:** Look up one more cultural example (e.g. al-Bīrūnī's magic squares, or the Jain *Pārśvanātha* yantra). Bring an image and a verified magic constant.
- **One tier down:** Focus on one comparison: how is the Lo Shu the same as our Day-1 square? Trace the rotation that takes one into the other.

Teacher notes

- Don't fall into "ancient Indians invented this first" framing. The Lo Shu is older than any dated Indian magic square we have. Nārāyaṇa's contribution is the *systematic method*, not the *first discovery*.
- The Dürer corner being a date (1514) often delights students. Use that delight to motivate the project.
- Some students will see the Khajuraho ~10th-c. dating and conclude it predates Lo Shu. That is incorrect. The Lo Shu has earlier documented evidence (Han dynasty texts). Be clear about which dates are *attestations* and which are *legends*.

Citation(s) used in this lesson

- Lo Shu: *Da Dai Liji* (Chinese, Han dynasty, ~5th–1st c. BCE) — earliest documented Chinese mention.
- Dürer, *Melencolia I* (1514) — material artefact.
- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14, 1356 CE — for context on Indian systematic treatment.

Day 9 — Sudoku Connection + Project Session

Module: Magic Squares (*Bhadragāṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) distinguish a Sudoku puzzle from a magic square (Sudoku constrains *placement* but not *sums*), and (b) make substantial progress on their module project.

Materials

- Sample 9×9 Sudoku puzzle (one per pair, easy difficulty)
- Project materials (graph paper, markers, etc.)

Vocabulary introduced today

- *Sudoku* — a 9×9 number-placement puzzle (Japanese name, from Japanese 数独). Standard rules: each row, each column, and each 3×3 block contains the digits 1–9 exactly once.
- *constraint satisfaction* — a problem type where you must place objects in a structure so that all “constraints” (rules) are satisfied. Magic squares and Sudoku are both constraint-satisfaction problems.

Lesson flow

Warm-up (5 min)

- “*Many of you have seen Sudoku. Is it a magic square?*” — take 3 student responses.
- Hold the answer. Promise: “we’ll settle this in 10 minutes.”

Core (15 min)

1. **The rules side-by-side.** Write on the board:

Property	Magic Square (3×3 , 1–9)	Sudoku (3×3 within 9×9)
Uses digits 1–9?	yes — once each	yes — once each within each 3×3 block
Each row uses each digit once?	no (a row has just 3 distinct values from 1–9)	yes (each row uses 1–9)
Each column uses each digit once?	no	yes
Diagonals sum to a constant?	yes	not required
Rows sum to a constant?	yes (= 15)	not enforced
Columns sum to a constant?	yes	not enforced

“Magic squares constrain sums . Sudoku constrains placement . Different constraints — different puzzles.”

2. **Both are constraint-satisfaction problems.** Each gives you a structure (9 cells in 3×3 , or 81 cells in 9×9) and a set of rules. You search for placements that satisfy all rules. Computers solve these the same way.
3. **A bridge example.** It is possible to have a 9×9 grid that is *both* a Sudoku AND a magic square (every row, column, *and* 3×3 block sums to 45 — which works because rows/columns automatically sum to $1+2+\dots+9 = 45$). Such squares exist; they are rare. Show one if you have time. (*Not required at Middle band.*)

Activity (20 min) — Project work

- Pairs work on their project. Teacher circulates.
- Required checkpoints by end of class:
 - The magic square is fully constructed (5×5 or larger).
 - Magic constant verified for at least 3 rows, 3 columns, both diagonals.
 - Citation is in place (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14, 1356 CE).
 - At least one paragraph (or one slide of a video) of explanation drafted.
- The teacher does a 1-minute review with each pair to flag any errors before Day 10.

Wrap-up (5 min)

- Quick poll: “On a scale of 1–5, how ready do you feel for Day 10’s share?” Show of fingers.
- For 1–2 ratings: teacher schedules a 5-min meeting with that pair before Day 10.

Student-facing content

Sudoku ≠ magic square.

	Magic square (3×3, 1–9)	Sudoku 3×3 block
Cells	9 distinct values from 1–9	9 distinct values from 1–9
Rows sum to a constant?	Yes (15)	No requirement
Each row has all 9 digits?	No (only 3 cells)	Yes (in 9×9 Sudoku)
Diagonals constrained?	Sums	Not required

Both are *constraint-satisfaction* problems. Different constraints — different puzzles.
Sudoku appeared in modern form in the 1970s–80s (popularised in Japan from 1986).
Magic squares are at least 2500 years old (Lo Shu).

Homework

- Finish anything not finished in class. Tomorrow is the share, not the start.
- Time yourself: how fast can you construct a 5×5 magic square now, versus on Day 5?

Differentiation

- **One tier up:** Find or construct a 9×9 grid that is both a Sudoku AND a magic square (every row/column/3×3 block sums to 45). Hint: rows and columns of a Sudoku already sum to 45. You only need to also make every 3×3 block sum to 45.
- **One tier down:** Use the class time to construct one more 5×5 magic square as practice for the summative. Speed is not the goal; correctness is.

Teacher notes

- The Sudoku discussion is short on purpose. The point is to draw a clean distinction, not to teach Sudoku.
- Spend the project-work time circulating. Most errors are caught at this stage.

- Some students will worry about the “design-your-own” requirement. Remind them: choosing the size, the starting number, and the presentation format are all design choices. They are not expected to invent a new algorithm.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — paraphrased in projects.

Day 10 — Summative + Share

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) complete the summative quiz, and (b) present their original magic-square project to the class.

Materials

- Printed summative quiz (one per student) — `quizzes/summative.md`
- Project artefacts (posters, devices for videos, written exposition)
- Stopwatch (for 2-minute presentation timer)

Lesson flow

Summative quiz (15 min)

- Closed notebook. Students complete `quizzes/summative.md` individually.
- 15 min total: 3 min reading, 12 min solving.

Presentations (25 min)

- Each pair gets **2 minutes** to present + 1 minute Q&A.
- With ~10–12 pairs in a typical class, this fits in 25 min.
- Order: teacher draws pair names randomly.

What each presentation must show:

- The constructed magic square (size, magic constant verified).
- Citation of the source (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, 1356 CE).
- One sentence on *why this particular design* — what choice did the pair make?

Wrap-up (5 min)

- One question per student on a slip of paper: “*What is one thing you’ll remember about magic squares one year from now?*”
- Collect. Read at random — 2–3 aloud.
- Close with: “*You all now know an algorithm older than most countries’ modern names. Use it. Show someone.*”

Student-facing content

Today is the share.

Bring: - Your project artefact (poster / video / written exposition). - Your two-minute presentation prepared. - Your full attention for your classmates' presentations — you will give feedback.

Reminder: **Verify before you present.** Sum at least 3 rows, 3 columns, and both diagonals out loud (or in your head). If they don't match the magic constant, fix the error *before* you stand up.

Homework — none

The module ends here. The project artefact stays with the school until the end of term as part of the module's portfolio. Reflection slips are collected.

Differentiation

- **One tier up:** Pairs who finish early help check others' projects against the rubric (specifically, the “method demonstrated correctly” criterion).
- **One tier down:** A pair that didn't quite finish presents a smaller artefact (e.g. 3×3 instead of 5×5). They still meet the citation and verification requirements.

Teacher notes

- Keep the presentations tight. 2 minutes is short on purpose — encourages preparation.
- The Q&A minute is for *one* clarifying question, not a defence. Cap it strictly.
- Score against `assessments/rubric.md` during the share — don't try to score after. Live scoring keeps things fast.
- For students who freeze: have a partner take over. The presentation is paired; both partners speak counts towards the rubric.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — referenced in project artefacts.

Writing Prompts

Writing Prompts — Module 14 Magic Squares (Middle)

Six prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

1. (Day 1, end of class, 3 min) When you tried to build the 3×3 magic square by trial and error, what was going through your head? Did you have a strategy, or were you just guessing? Be specific.

2. (Day 7, end of class, 5 min) You spent three days using Nārāyaṇa’s *turagagati* algorithm before learning who Nārāyaṇa was. Did learning about him change how you feel about the method? (Either answer — yes, no, somewhere in between — is fine, but explain it.)

Explanatory prompts

3. (Day 4 or homework, ~120 words) Explain the *turagagati* method to a Grade 5 student who has never seen it. Use no more than 7 sentences. Include one specific example.

4. (Day 6 or homework, ~120 words) Explain *why* the Khajuraho 4×4 square is called “panel-magic.” Use the word **constraint** in your answer.

Argumentative prompt

5. (Day 8 or homework, 1 paragraph) Some sources call magic squares “Indian magic” or “Chinese magic” depending on the culture. Others insist that magic squares appeared independently in many cultures, and no single culture “owns” them. Pick a position and defend it in 5–7 sentences. Use what you know about Lo Shu, Khajuraho, Nārāyaṇa, and Dürer.

Creative prompt

6. (Day 7 or Day 8, optional, 1 page) Write a short dialogue (50–150 words) between Nārāyaṇa Paṇḍita (1356 CE) and Albrecht Dürer (1514 CE), meeting across time. They show each other their magic squares. Let them disagree on something — maybe about *why* they care about these patterns. Don’t have them magically agree.

Prompt-to-band notes

- For **Primary** band, prompts 1, 3, and 6 work as drawings or oral responses.
- For **Senior** band, prompt 5 deserves a full essay treatment with sources cited from `sources.md`.

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction.
- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the rubric — does the writing acknowledge the other side, give a specific example, reach a clear position?
- Prompt 6: no grading. Read the best ones aloud (with permission).

Homework

Homework — Module 14 Magic Squares (Middle)

Day 1 → Day 2

Task: Copy the 3×3 magic square from class. Beside it, draw three rotations (90° , 180° , 270°). Verify one row of each.

Time: 10 min.

Day 2 → Day 3

Task: Verify the magic constant formula $S = n(n^2 + 1)/2$ for $n = 8$ and $n = 9$. Show your arithmetic. (Answers: 260 and 369.)

Time: 8 min.

Day 3 → Day 4

Task: Take the 3×3 magic square and add 10 to every entry. The square becomes one with numbers 11–19 instead of 1–9. What is the new magic constant? Verify one row.

Time: 10 min.

Day 4 → Day 5

Task: Construct three 5×5 magic squares from scratch using the *turagagati* method. Verify the magic constant of each (should be 65). Yes, they should all be the same square — that's part of the discovery.

Time: 20 min.

Day 5 → Day 6

Task: Construct a 7×7 magic square using the stair-step method. Magic constant should be 175. Bring it tomorrow for spot-checks.

Time: 25 min.

Day 6 → Day 7

Task: Pick any 2×2 panel of the Khajuraho square (including one that wraps the edge). Verify the sum is 34. Write down both the panel cells and the sum.

Time: 5 min.

Day 7 → Day 8

Task: Polish your in-class reflection paragraph on Nārāyaṇa. Add one more specific example from your own construction work.

Time: 15 min.

Day 8 → Day 9

Task: Submit your project plan (4–6 sentences): format, target audience, the magic square you'll construct, what is unique about your design.

Time: 15 min.

Day 9 → Day 10

Task: Finalise project artefact and rehearse the 2-minute presentation aloud at least once. Time yourself.

Time: 30 min (cap).